Learning Individual Comfort: Personalized Proxemic Greeting for a Simulated Social Robot via Reinforcement Learning

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I. Problem Addressed

Social robots optimised for task completion routinely ignore that individuals differ: the interpersonal distance and greeting intensity one person finds welcoming, another finds intrusive. Reinforcement learning has been shown to adapt robot behaviour to users [1], [2], [3], [4], but those evaluations are small (N=8–28), hardware-bound, and rarely separate adaptation — learning a good behaviour — from personalization — learning a different behaviour for each person.

This work addresses that distinction: does a social robot actually need to personalize, or is one well-trained policy enough for everyone? Personalization is expensive — it requires per-user interaction data — so answering this demands comparison against a trained generalist, not merely a scripted robot.

II. Approach

We implement greeting/proxemics as a discrete MDP in simulation. State is the sensed distance bin (Hall's proxemic zones) crossed with commanded greeting intensity ($4 \times 3 = 12$ states); actions are {approach, withdraw, intensify, soften, hold}. Reward uses social signals only — $+1$ approval, -2 if the person disengages, -0.05 per step — so the robot never observes the preference it must satisfy. Learning is tabular Q-learning: interpretable, and 12 states need no function approximation and no GPU.

Individual difference is modelled by a UserProfile: three simulated people (reserved, neutral, sociable) with different hidden comfortable distances and preferred intensities, placed in mutually exclusive distance bins so a policy optimal for one is wrong for another. We compare five conditions: scripted; one policy trained across the mixed population; the strongest coherent single policy; personalized per user; and warm-started from other users then adapted (leave-one-out).

All work is in simulation: without access to a physical humanoid, hardware trials and a subjective human study are explicitly out of scope and remain future work. The learning code addresses an abstract robot interface rather than any specific platform, so the identical loop would run against hardware unchanged.

III. Expected Outcome

The implementation is complete; preliminary results follow (8 seeds, 6000 episodes, mean reward per interaction):

Condition	Reserved	Neutral	Sociable
Scripted	−1.26	2.03	−0.75
One policy, mixed	6.45	−0.75	−0.75
Best single policy	**11.81**	−0.75	−0.75
Personalized	**11.83**	**10.31**	**8.34**

The result is unambiguous: the best single non-personalized policy is optimal for one user (11.81) and worse than doing nothing for the other two (-0.75), while personalization gains $+11.06$ and $+9.09$ there. It is not a refinement; it is the difference between a working greeting and a failed one.

We also report a negative result: warm-starting from other users did not help, merely matching from-scratch learning at equal budget and faring worse at a small one. We attribute this to a boundary condition on the transfer idea in PbARL [1], which works because task competence and user preference are separable — the robot must still complete the task, only differently. In proxemics the task is the preference, so another person's optimum carries no information about this one's. This is a useful limit rather than a failure, and the result most worth discussing.

The demonstration presents learning curves, the per-user learned policies side by side, and the comparison above, executed live. The implementation is publicly available[1] with a test suite; every reported number is reproducible from a fixed seed.

References

[1] R. Wang et al., "Personalization in HRI through preference-based action representation learning," IEEE ICRA, 2025.
[2] D. C. Tozadore et al., "Towards adaptation and personalization in task based on HRI," LARS/SBR/WRE, 2018.
[3] A. Gonzalez-Santocildes et al., "Adaptive robot behavior based on human comfort using RL," IEEE Access, 12, 2024.
[4] E. Zedda et al., "Adaptive behaviour-based strategy for SARs with MCI older adults," arXiv:2305.01492, 2023.

[1] https://github.com/nourgaser/nao-adaptive-rl